Artificial Intelligence

Uncertainty

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1 From Logic to Probability Theory

Consider:

$$Toothacke \implies Cavity$$

Not all patients with toothaches have cavities. How about:

$$Toothache \implies Cavity \lor GumProblem \lor Abscess \ldots$$

We'd need large list after the dots. Try causal direction:

$$Cavity \implies Toothache$$

But not all cavities hurt. Logic fails to deal with complex domains like medical diagonosis due to:

- **Exhaustivity** (laziness). Too many antecedents or consequents to list.
- **Theoretical ignorance**. Rare to have complete logical theory of nontrivial domain.
- **Practical ignorance**. Even with a complete domain theory, hard to measure all the necessary inputs (medical tests, etc.)

We need to deal with uncertain knowledge, with **degrees of belief**. For that, we use probability theory.

2 What do probabilities mean?

3 What do probabilities mean?

Proposition	Agent 1's belief	Agent 2 bets	Agent 1 bets	Agent 1 payoffs for each outcome a, b	$a, \neg b$	$\neg a, b$	$\neg a, \neg b$
a	0.4	\$4 on a	\$6 on $\neg a$	–\$6	–\$6	\$4	\$4
b	0.3	\$3 on b	\$7 on $\neg b$	–\$7	\$3	–\$7	\$3
$a \vee b$	0.8	\$2 on $\neg(a \vee b)$	\$8 on $a \vee b$	\$2	\$2	\$2	–\$8
				–\$11	–\$1	–\$1	–\$1

All I see is the `code`.

```
this_is = a_python_code_block
if it_is_not_processed:
    it_is_likely_within_a_column_env
    if not_in_colmn_env:
        there_is_another_problem
```

4 Probabilities are about our uncertain knowledge.

"80% chance (probability 0.8) patient with a toothache has a cavity."

- Out of all the situations that are indistinguishable from the current situation **as far as our knowledge goes**, the patient will have a cavity in 80% of them.
- Belief could come from statistical data, or domain theory, or combination of sources.

  There is no uncertainty in the actual world. The patient either has a cavity or does not. Probabilities refer to our knowledge of the world state, not the actual world state.[fn]

If our knowledge changes, e.g., we find out patient has history of gum disease, we make a different statement. Here is a .

"Given patient has a history of gum disease and a toothache, 40% chance patient has gum disease."

[fn]: Footnote [fn2]: `https://this/isn't/real`

5 Decision-Theoretic Agents

Decision theory = probability theory + utility theory

function DT-AGENT(*percept*) **returns** an *action*
  persistent: *belief_state*, probabilistic beliefs about the current state of the world
    action, the agent's action

  update *belief_state* based on *action* and *percept*
  calculate outcome probabilities for actions,
    given action descriptions and current *belief_state*
  select *action* with highest expected utility
    given probabilities of outcomes and utility information
  return *action*

Each action, a, leads to a probability distribution over outcomes, or "result states":

$$P(\text{RESULT}(a) = s') = \sum_{s \in S} P(s)P(s' \mid s, a)$$

And each outcome state has a utility. A rational decision-theoretic agent chooses the action that maximize expected utility, that is:

$$\operatorname*{argmax}_a \sum_{s' \in S} P(\text{RESULT}(a) = s')U(s')$$

6 Propositions and Events

An **event**, which we denote with ϕ here, is a set of possible worlds, some subset of Ω, or set of ω, e.g., the event "doubles" contains the 6 boxed elements ω:

$\boxed{(1,1)}$	(2, 1)	(3, 1)	(4, 1)	(5, 1)	(6, 1)
(1, 2)	$\boxed{(2,2)}$	(3, 2)	(4, 2)	(5, 2)	(6, 2)
(1, 3)	(2, 3)	$\boxed{(3,3)}$	(4, 3)	(5, 3)	(6, 3)
(1, 4)	(2, 4)	(3, 4)	$\boxed{(4,4)}$	(5, 4)	(6, 4)
(1, 5)	(2, 5)	(3, 5)	(4, 5)	$\boxed{(5,5)}$	(6, 5)
(1, 6)	(2, 6)	(3, 6)	(4, 6)	(5, 6)	$\boxed{(6,6)}$

7 Analysis of Medical Screening Example

$$
\begin{aligned}
P(C = 1) &= \frac{1}{100} \\
P(C = 0) &= \frac{99}{100} \\
P(T = 1 \mid C = 1) &= \frac{90}{100} \\
P(T = 0 \mid C = 1) &= \frac{10}{100} \\
P(T = 1 \mid C = 0) &= \frac{3}{100} \\
P(T = 0 \mid C = 0) &= \frac{97}{100}
\end{aligned}
$$

If we screen someone, probability that they test positive (notice use of product rule: $P(a, b) = P(a \mid b)P(b)$ and rule 2nd Kolmogorov axiom $P(x \vee y) = P(x) + P(y) - P(x \wedge y)$):

$$
\begin{aligned}
P(T = 1) &= P(T = 1 \mid C = 0)P(C = 0) + P(T = 1 \mid C = 1)P(C = 1) \\
&= \frac{3}{100} \times \frac{99}{100} + \frac{90}{100} \times \frac{1}{100} \\
&= \frac{387}{10,000} \\
&= .0387
\end{aligned}
$$

If someone tests positive, probability they have cancer :